Towards a Framework for Interactive Extraction of Semantic Information from Unstructured Clinical Documents

Lucas Kulla[1,2,3,†], Leon Remke[1,2,†], Sthuthi Sadananda[1,2,†], David Schwenke[1,2], Simon Kreutzfelt[1], Eva Krieghoff-Henning[1], Eva Reisinger[1], Stefan Fröhling[1], Klaus Maier-Hein[3], Marco Nolden[3,‡] and Keno März[1,2,‡]

[1]*National Center for Tumor Diseases (NCT), NCT Heidelberg, a partnership between DKFZ and University Hospital Heidelberg, Germany.*

[2]*NCT Digital Oncology Program*

[3]*Division of Medical Computing, Deutsches Krebsforschungszentrum (DKFZ) Heidelberg*

Abstract

In Germany, roughly 410 000 unstructured clinical documents are generated daily [1]. These documents contain rich information that is only partially reflected in structured data, and thus not accessible for primary treatment or secondary use in research. Manual extraction is time-consuming and error-prone, limiting scalability [2]. Existing information extraction (IE) pipelines are often tailored to narrow use cases and difficult to generalise, or lack explainability of results, effectively operating as black-box systems [3]. This work-in-progress contribution outlines an interactive semantic framework for IE from unstructured clinical documents developed at the Digital Oncology Program (NCT Heidelberg) . The framework combines automated first-pass extraction using natural language processing (NLP) and large / vision language models with semantic normalisation and mapping to user-defined target schemas and clinical terminologies. A dedicated web-based interface enables human-in-the-loop curation by presenting extracted values alongside highlighted source spans, ensuring source-to-target traceability. As part of the NCT/DKTK/DKFZ MASTER programme, the first prospective evaluation will compare manual documentation with framework-assisted workflows. This will assess the quality, efficiency, transparency and user acceptance of extractions, with a focus on tightly coupling AI-based extraction with semantic modelling and expert interaction. The framework aims to provide a generalisable approach to transparent information extraction in oncology.

Keywords

clinical NLP, language model, informaton extraction, oncology, clinical application

1. Introduction

Clinical routine generates large volumes of free-text documents containing essential diagnostic and therapeutic information [1]. Only limited subsets are transferred to structured systems, creating data gaps that impact research and secondary use. Manual documentation is resource-intensive and error-prone. Existing extraction pipelines are typically tailored to narrow use cases, difficult to generalise, and are insufficiently transparent for domain experts. Advances in large language models (LLM), vision-language models (VLM), retrieval-augmented generation, and knowledge graphs can enable semantically grounded extraction from unstructured clinical text [2]. However, clinical adoption requires workflows that support expert oversight and provenance tracking. This work develops an interactive semantic framework addressing these requirements. The objectives are to (1) identify extraction and mapping steps that generalise across heterogeneous oncology use cases, (2) integrate domain-expert input at critical workflow stages, and (3) provide comprehensive source-to-target traceability. The central hypothesis is that an interactive, semantically grounded IE yields more robust and transparent results than fully automated black-box approaches.

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[†]These authors contributed equally to this work

[‡]These authors share last authorship

lucas.kulla@dkfz.de (L. Kulla)

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2. Framework Overview & Use Cases

The framework comprises three stages: (1) automated first-pass extraction, (2) semantic transformation and mapping, and (3) interactive knowledge acquisition. In the first stage, unstructured documents are processed using OCR and VLM/LLM components when required. Classical NLP and LLM-based methods propose candidate entities and attributes according to a configurable target schema (e.g., oncological trial registry requirements). In the second stage, extracted candidates are semantically normalised and mapped to target schemas while provenance metadata are captured at each transformation step. In the third stage, structured candidates are presented in a web-based interface alongside highlighted source spans in the original document. Clinical experts and documentation staff can validate results. All edits are logged for iterative refinement of the process. The framework is deployed across multiple oncology-related scenarios at the National Center for Tumor Diseases, including the NCT/DKTK/DKFZ MASTER programme and selected pathology workflows. A prospective study will compare manual and framework-assisted documentation with respect to extraction quality (F1-score against curated gold standards), efficiency (time per document and field), perceived transparency, workload, and qualitative assessments of traceability and FAIRness.

3. Status and Expected Contributions

Current work focuses on applying the prototype across use cases involving heterogeneous schemata and preparing a prospective validation. The interactive framework aims to demonstrate measurable improvements in documentation quality and process efficiency while supporting transparent integration of AI-based models into clinical environments. In the long term, it is expected to reduce manual data entry, improve data quality, and support translational oncology research.

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Declaration on Generative AI

During the preparation of this work, the author(s) used GPT-5.2 in order to perform grammar and spelling checks and to assist in text condensation. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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